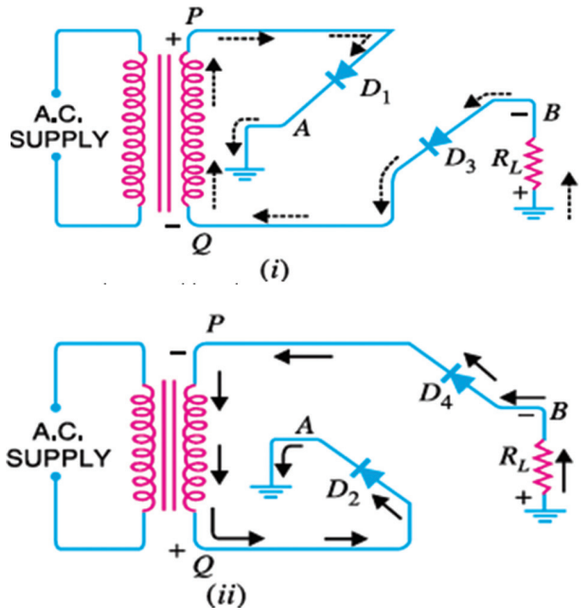


Fig 10.5 *Full Wave Bridge Rectifier*

B through the load.ie in the same direction as for the positive half-cycle. Therefore d.c output is obtained across load R_L

The peak inverse voltage (piv) of each diode is equal to the maximum secondary voltage of transformer.

10.3 FILTER CIRCUITS

Generally a rectifier is required to produce pure d.c supply for using at various places in the electronic circuits. However, the output of a rectifier has pulsating character. ie it contains a.c and d.c components. The a.c. component is undesirable and must be kept away from the load. So a filter circuit is used to remove the a.c component.

> "A filter circuit is a device which removes the ac component of rectifier output but allows the d.c component to reach the load"

The most commonly used filter circuits are capacitor filter, choke input filter and capacitor input filter (or) π Filter.

10.4 ZENER DIODE

Zener Diode is a specially designed pn-junction diode. The symbol of the Zener diode in as shown in fig10.6. When forward biased, its characteristics are just those of ordinary diode. A Zener diode is always reverse connected, i.e. it is always reverse biased. When the reverse bias on a crystal diode is increased, a critical voltage

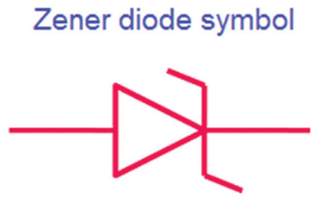


Fig 10.6 *Zener Diode*

called breakdown voltage is reached where the reverse current increase sharply to a high value. The breakdown region is the knee of the reverse characteristic as shown in fig 10.7. Therefore the breakdown voltage

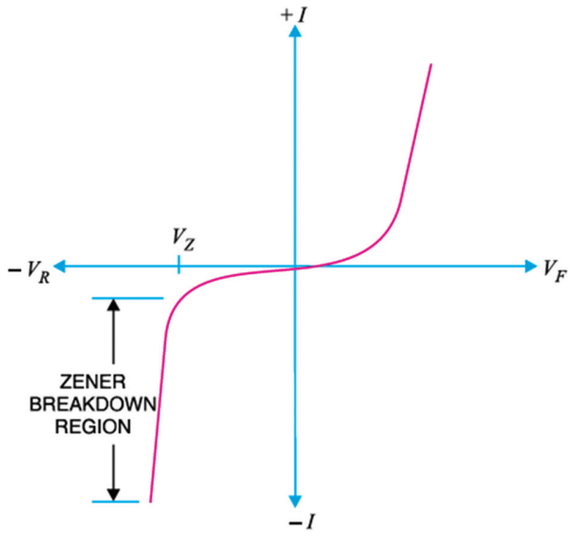


Fig 10.20 *Zener Diode Characteristics*

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